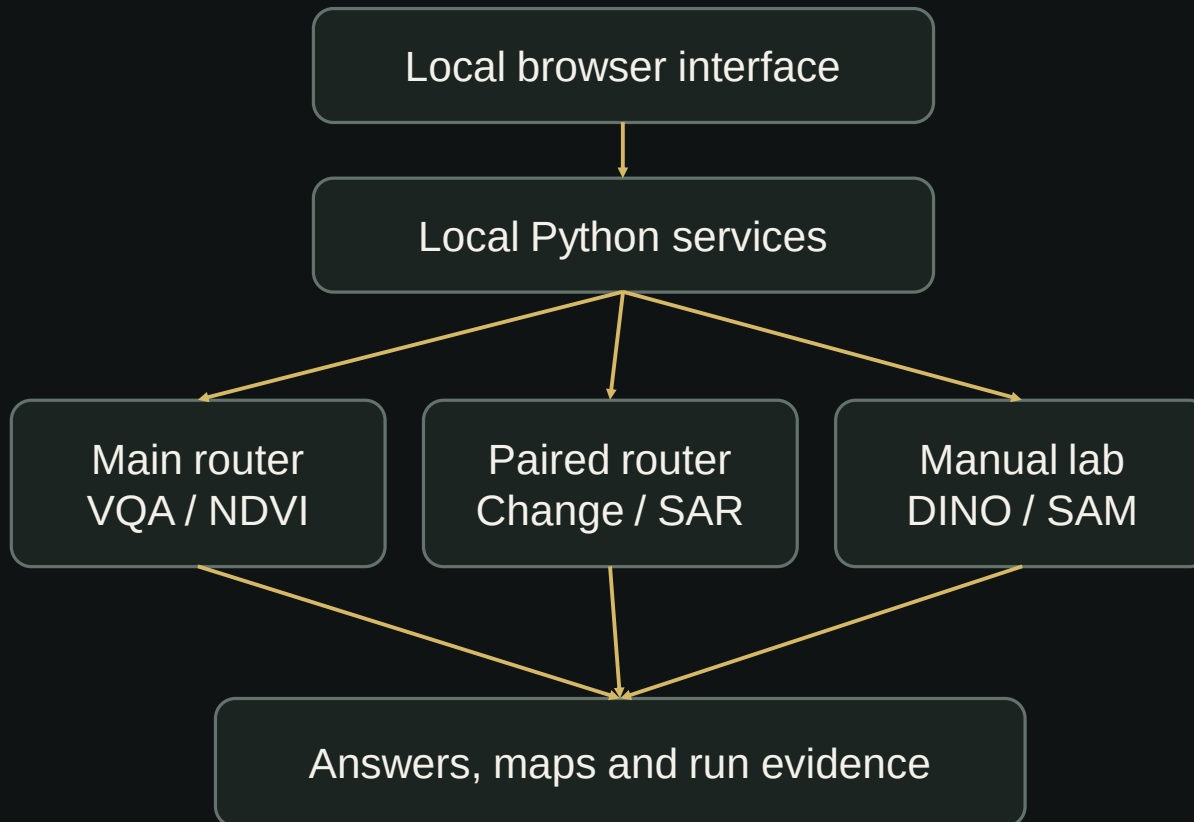


The whole app, at a glance

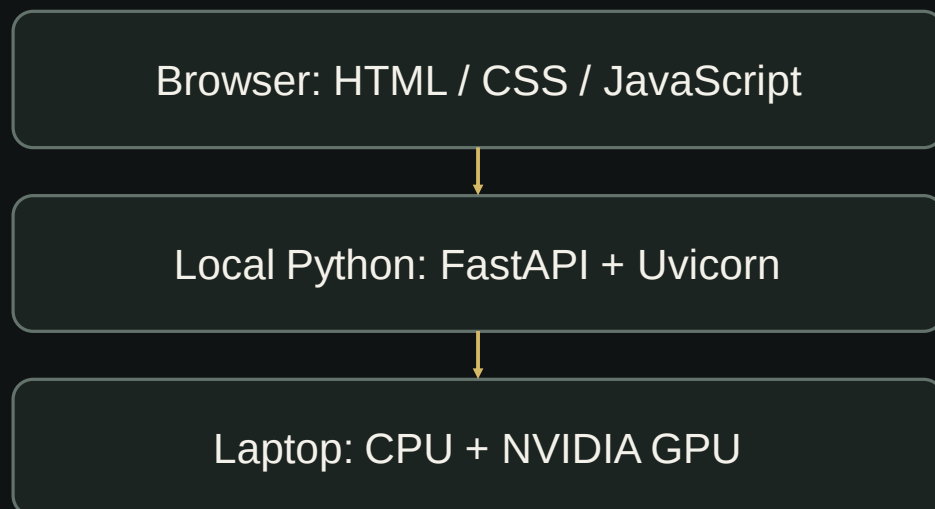
A question reaches the right tool. The tool returns an answer or visual evidence.



- The original demo stays separate from experimental model workflows.
- Every route has a defined input type and a limited supported task.
- A map is a prediction or calculation, with its method shown in run details.
- Future integration follows testing; grounding is not automatically routed today.

Local hosting and hardware

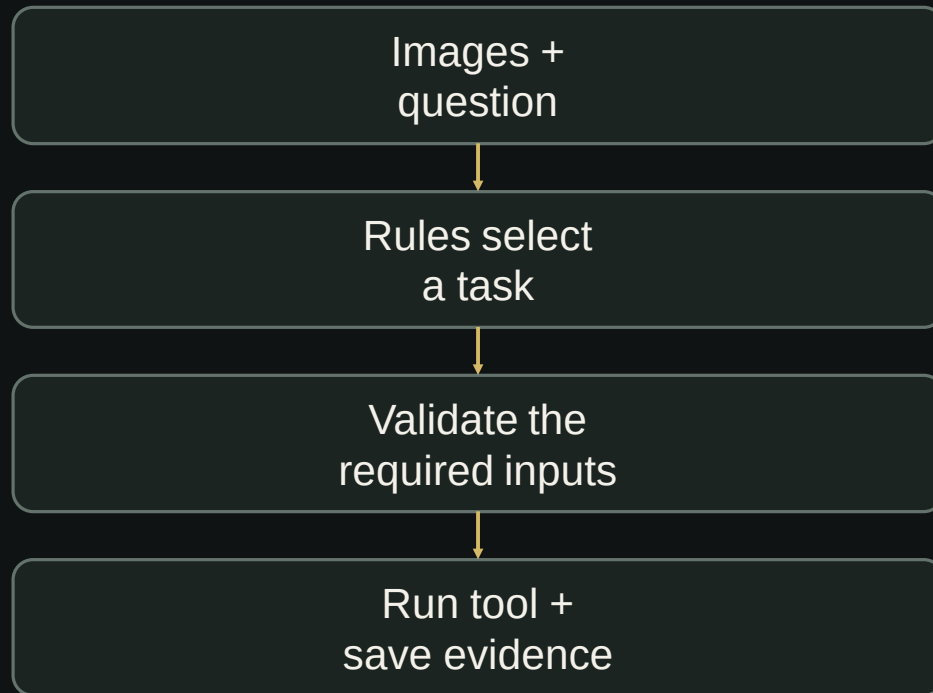
Everything runs on the laptop through browser pages and local Python services.



- localhost:8765 hosts the main app; /grounding opens its manual lab.
- localhost:8767 hosts the paired lab and connects to the main service.
- VQA uses the GPU. Paired inference uses the CPU in the current app.
- Grounding and SAM take turns on the GPU; models load from local storage.

Routing: choosing the right worker

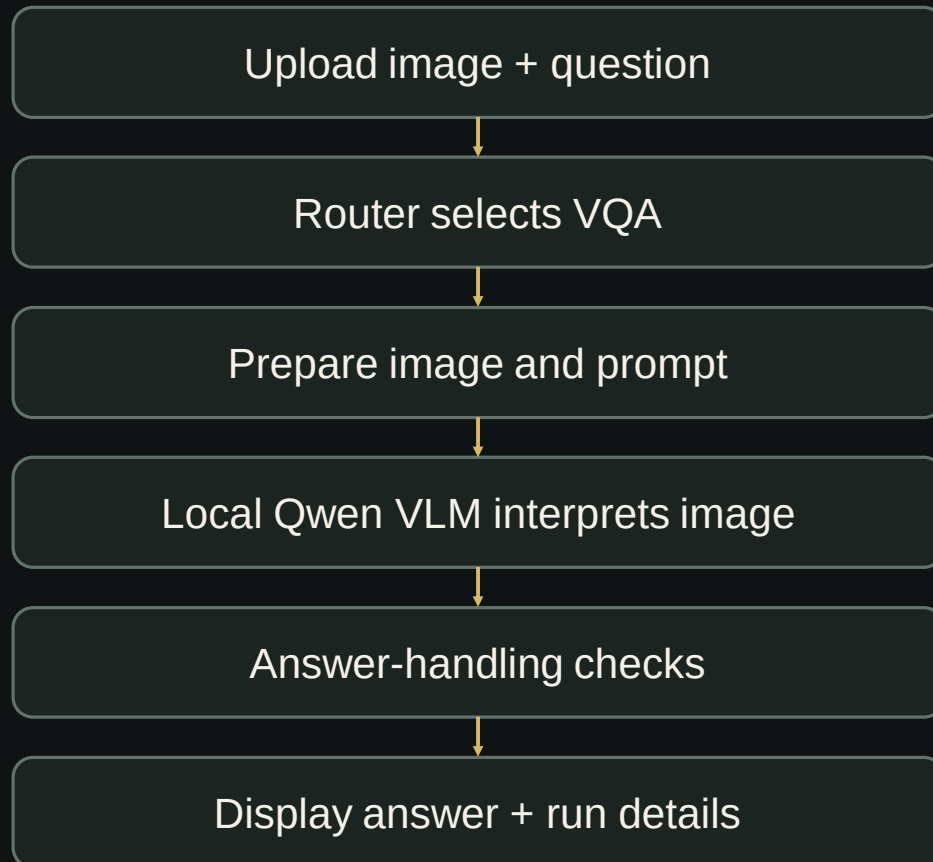
The controller reads the question and checks what images were supplied.



- One optical image: visual questions or calibrated NDVI calculations.
- Two optical dates: supported road and building change interpretation.
- Optical + radar: supported water mapping or scene classification.
- Ambiguous or incompatible requests stop with an explanation. Grounding stays manual.

VQA: asking a picture a question

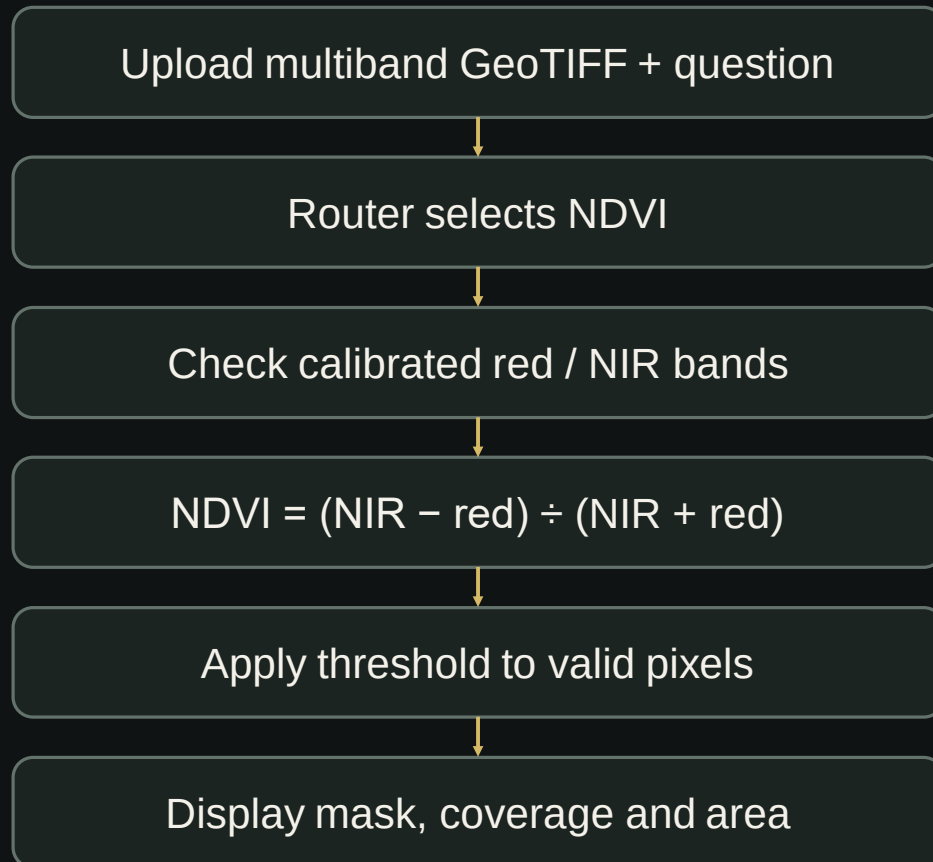
Example: “What is visible in this image?”



- Helps a user understand visible objects and the overall scene.
- Hugging Face Transformers and PyTorch run the local VLM.
- A PEFT / LoRA adapter supports the rural-or-urban classification path.
- Counts and descriptions can be wrong; they are not surveyed measurements.

NDVI: measuring a vegetation signal

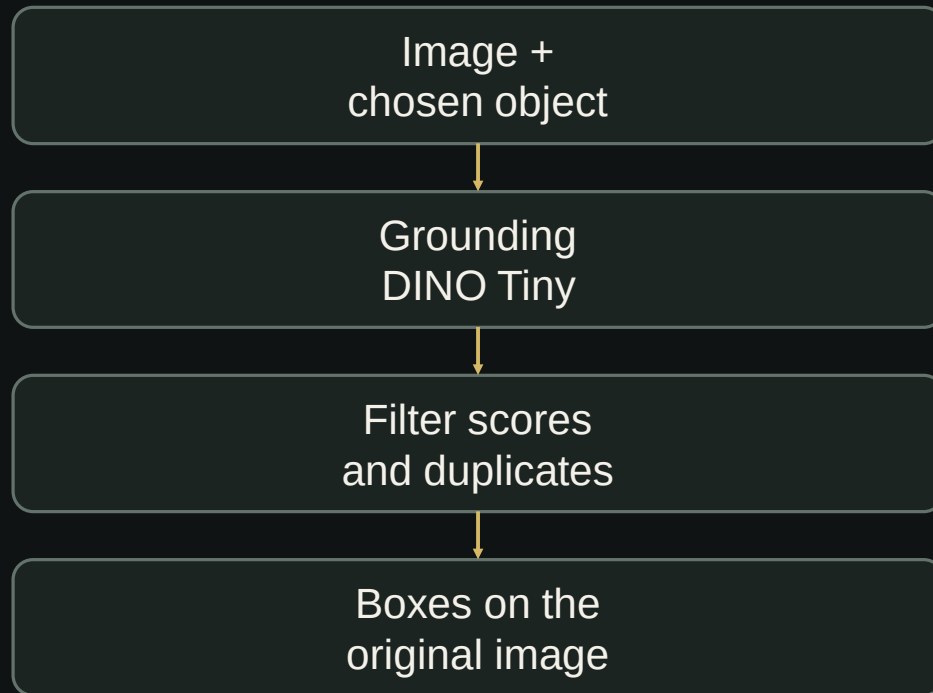
Example: “What percentage has NDVI above 0.5?”



- Rasterio reads band and map information; NumPy performs the calculation.
- Selected valid pixels ÷ all valid pixels gives threshold coverage.
- RGB, false-colour and mask views help explain the result.
- An ordinary RGB image lacks the near-infrared measurements this needs.

Grounding: finding the requested object

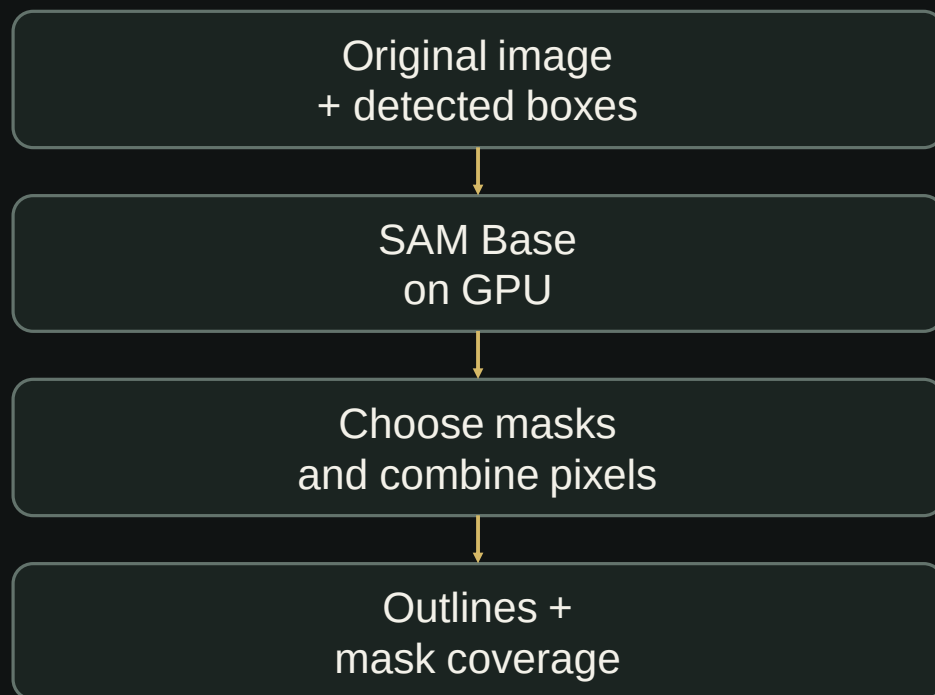
Example: choose “stadium” in the manual lab.



- Helps show where a requested object may be in the image.
- Transformers and PyTorch run IDEA-Research Grounding DINO Tiny.
- Lower score thresholds keep more candidates and more false alarms.
- Boxes use image pixels. Dense building detection remains a limitation.

Segmentation: tracing the object's pixels

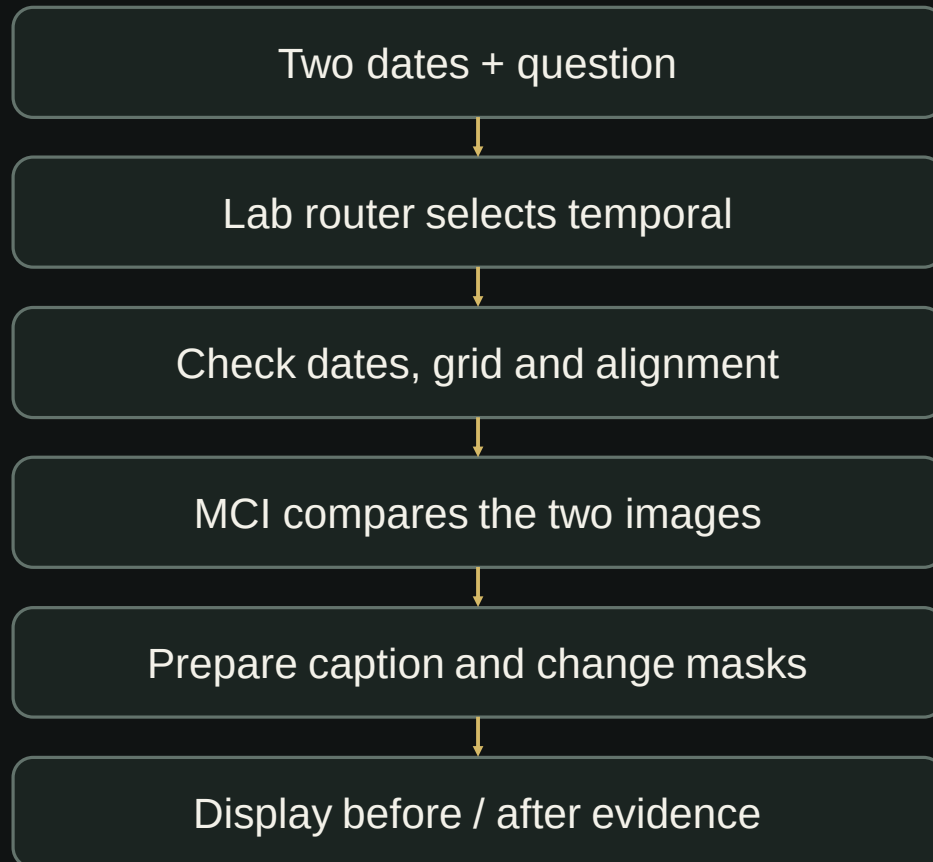
The grounding box tells SAM which region to outline.



- Helps inspect the shape of an object rather than only its rectangle.
- SAM uses the detected boxes as prompts for its pixel masks.
- Coverage = pixels in the combined mask ÷ pixels in the full image.
- A convincing outline can still be wrong if the detector chose the wrong object.

Temporal: comparing two dates

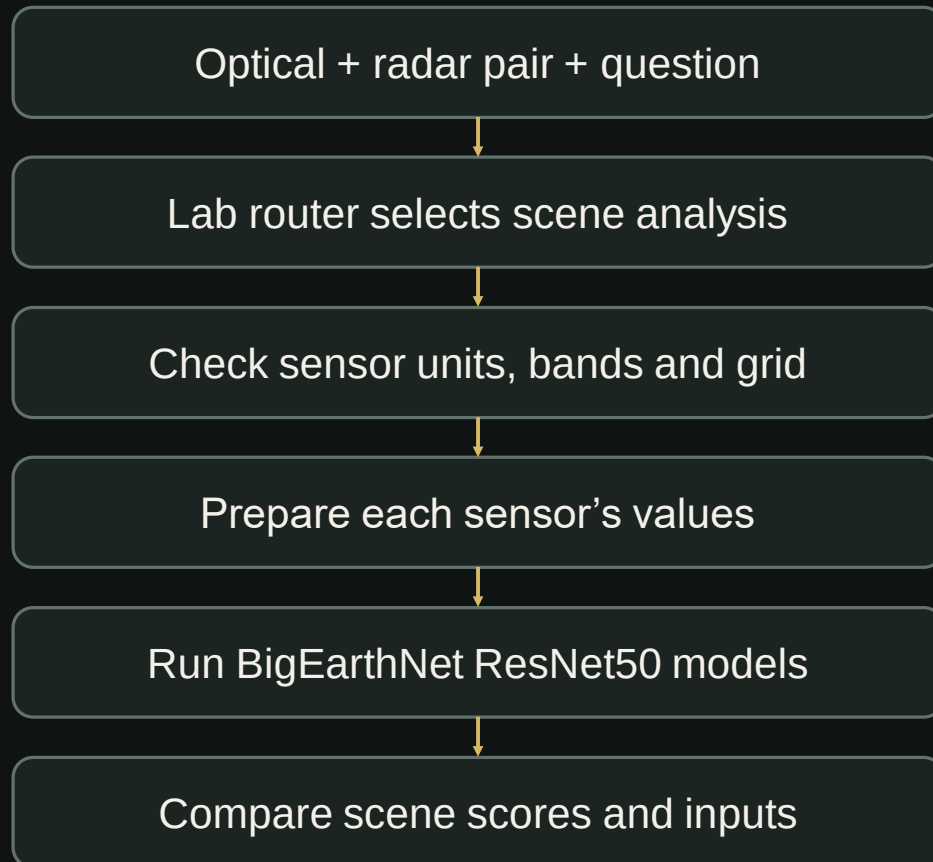
Example: “What changed between these two images?”



- Helps users inspect changes at the same location over time.
- OpenCV checks optical alignment; PyTorch runs the MCI model.
- The model paints predicted road and building changes and writes a caption.
- The caption is an interpretation; measured increase or decrease is not established.

Optical + SAR: understanding the scene

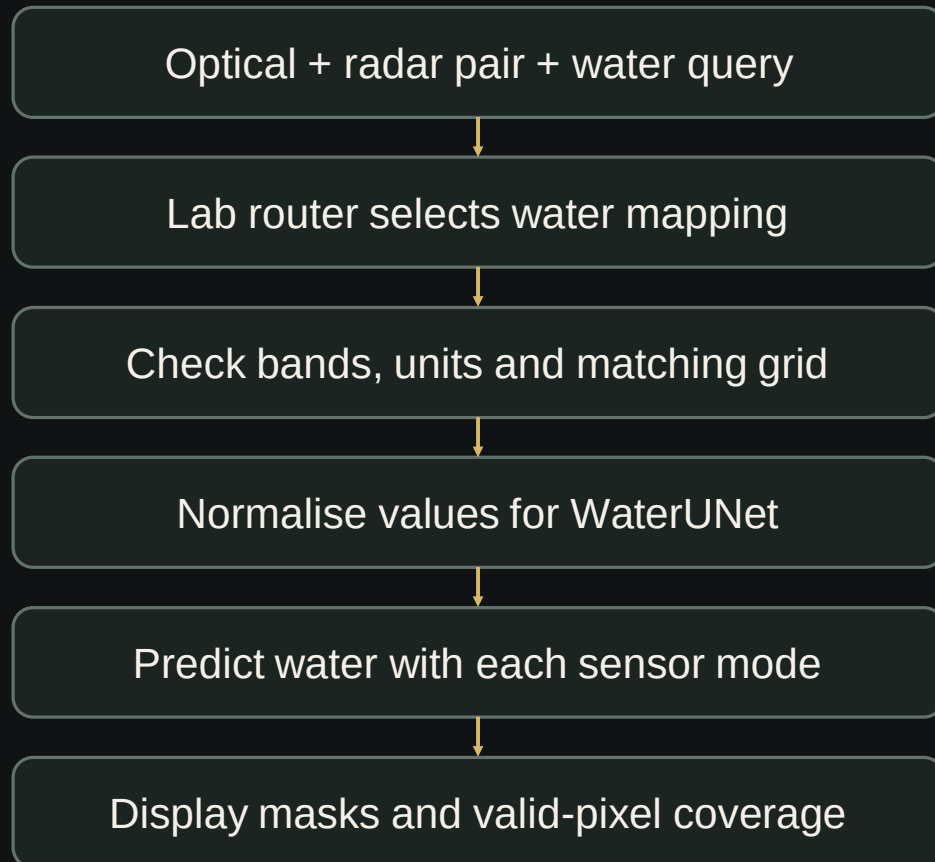
Two sensors describe the same geographic area in different ways.



- Optical imagery gives spectral detail; radar adds another sensor view.
- We compare radar-only, optical-only and joint scene predictions.
- Pretrained BigEarthNet models run on the CPU through PyTorch.
- These are scene categories, not object locations or guaranteed sensor improvements.

Water mapping: painting likely water

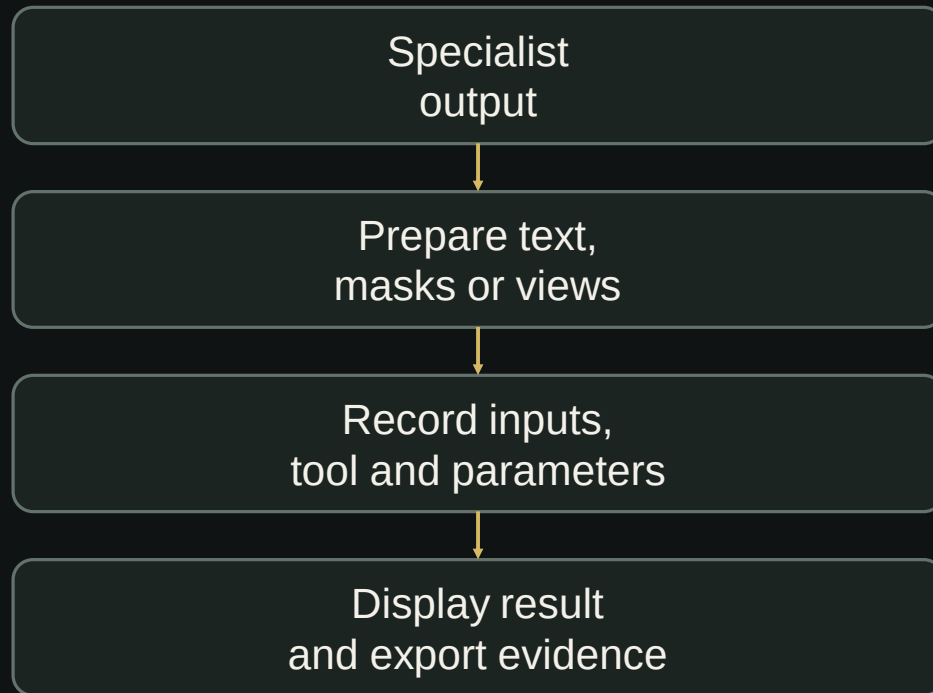
Example: “Map water using this optical and SAR pair.”



- We trained this small water model using real Sen1Floods11 image pairs.
- The same checkpoint runs radar-only, optical-only and joint predictions.
- Rasterio reads data; NumPy prepares it; PyTorch predicts water pixels.
- Coverage uses valid pixels. A single-date water map does not prove flooding.

Evidence: showing how we reached an answer

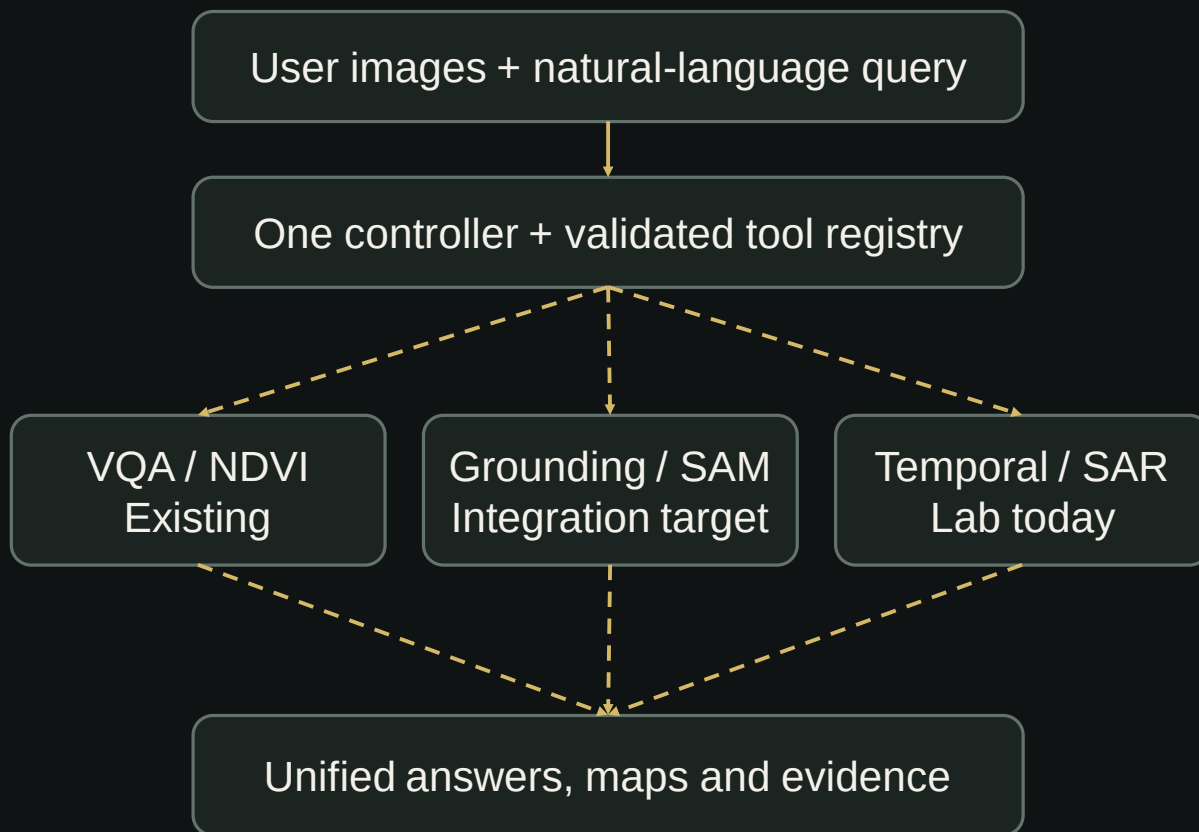
The user can inspect the result, the selected tool and the inputs behind it.



- Run details explain which specialist ran and which settings it used.
- Maps and overlays help users inspect the prediction or calculation.
- Paired runs export original inputs, results and produced views in a ZIP.
- Saved evaluation scores describe tested datasets, not every future image.

Our target: one tested system

Today's experiments become integrated features only after they pass the right tests.



- Plain image chat: image and prompt produce a text answer.
- Today: VQA / NDVI routing, a paired router and a separate manual grounding lab.
- Target: one controller connects specialists after input and performance tests.
- Before submission: strengthen benchmark evidence and document remaining limits.